

Chapter 1

Materials And Method

This chapter describes the experimental apparatus used in this work. First, a brief overview of the implementation of the Magneto-Optical Trap (MOT) for rubidium-87 atoms (^{87}Rb) is presented (this setup was already available prior to the present thesis work). The chapter then focuses on the work carried out during this thesis, namely the implementation of the Optical Dipole Trap setup employed to transfer and trap atoms from the MOT.

Although beyond the scope of the current work, it is worth noting that the long-term objective of this experiment is to confine the atoms within the core of a hollow-core fiber.

1.1 Magneto-optical trap

1.1.1 The Vacuum Chamber

1.1.2 The laser system

1.1.2.1 Master laser

1.1.2.2 Cooler and repumper lasers

1.2 Optical dipole trap

This section describes the implementation of the Optical Dipole Trap (ODT) setup used to transfer atoms from the MOT into a conservative optical potential. The ODT beam is guided through a Hollow-Core Photonic Crystal Fiber (HCPCF). As mentioned in Section 1.1, the HCPCF integrated into the experimental apparatus was already present prior to this thesis work. However, a test HCPCF was employed during the development of the setup in order to optimize the alignment procedure.

During the implementation of the system, dedicated diagnostic tools were also developed to investigate the near-field beam profile and improve the fiber-coupling alignment. The following subsections describe the ODT optical configuration and the experimental procedures adopted for the alignment and loading of the trap.

1.2.1 Optical Setup

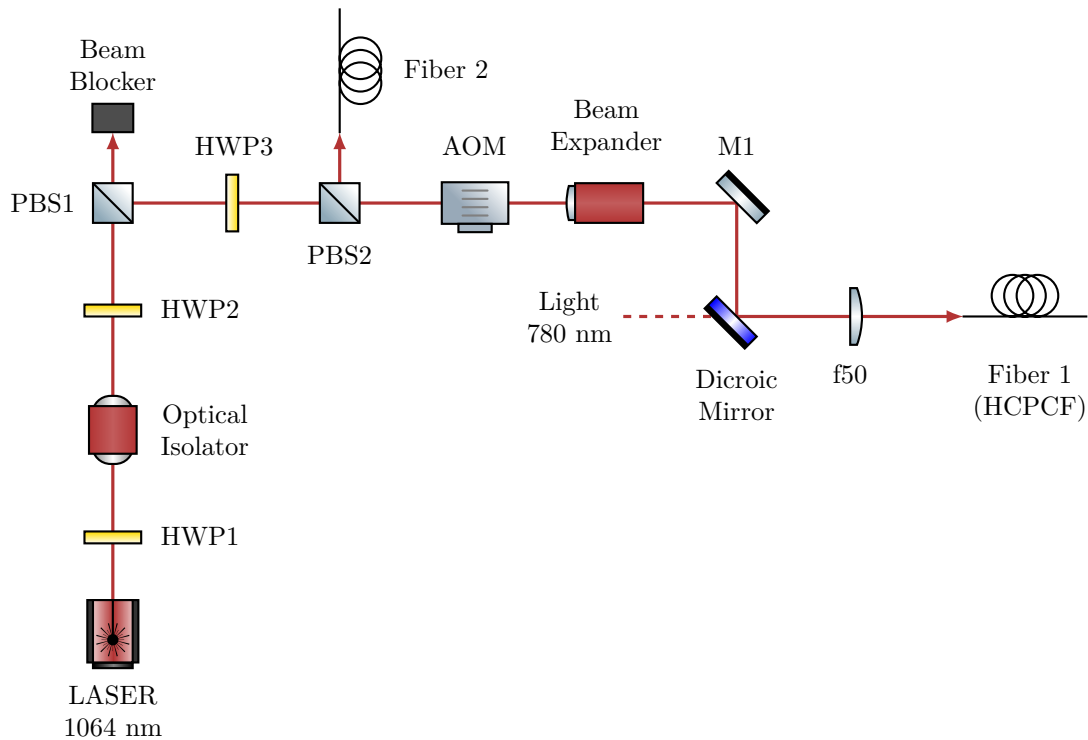


Figure 1.1: This is the caption

1.2.2 Near-Field Mode Characterization

1.2.2.1 Test Fiber

1.2.2.2 Chamber Fiber

1.2.3 MOT to ODT Loading Sequence